Unit 7

THERMODYNAMICS

Classical thermodynamics... is the only physical theory of universal physical theory of universal content which I am convinced... will never be overthrown.

Albert Einstein



Rudolf Clausius
(1822-1888)

Learning Objectives:

After studying this unit, students will be able to

- distinguish between the system and surroundings
- define closed, open and isolated systems
- distinguish between state and path functions
- describe the relation between internal energy, work and heat
- state four laws of thermodynamics
- correlate internal energy change (ΔU) and enthalpy change (ΔH)and their measurement.
- calculate enthalpy changes for various types of reactions
- apply Hess's law to calculate lattice energy of crystals
- define spontaneous and non spontaneous processes
- establish the relationship between thermodynamic state functions enthalpy (H), entropy (S) and Gibbs free energy (G)
- list the factors that determines spontaneity of the process
- establish the link between ΔG and spontaneity, and the relationship between ΔG^0 and equilibrium constant

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